

Nivarak - Complete Frontend UI Architecture & Component Mapping

Based on: Nivarak HLD + LLD v1.1 (June 2026)

Frontend Stack: Next.js 14 (App Router) · shadcn/ui · Tailwind CSS · TanStack Query · Zustand · React Hook Form + Zod · Recharts · TanStack Table

Location: nivarak/web/

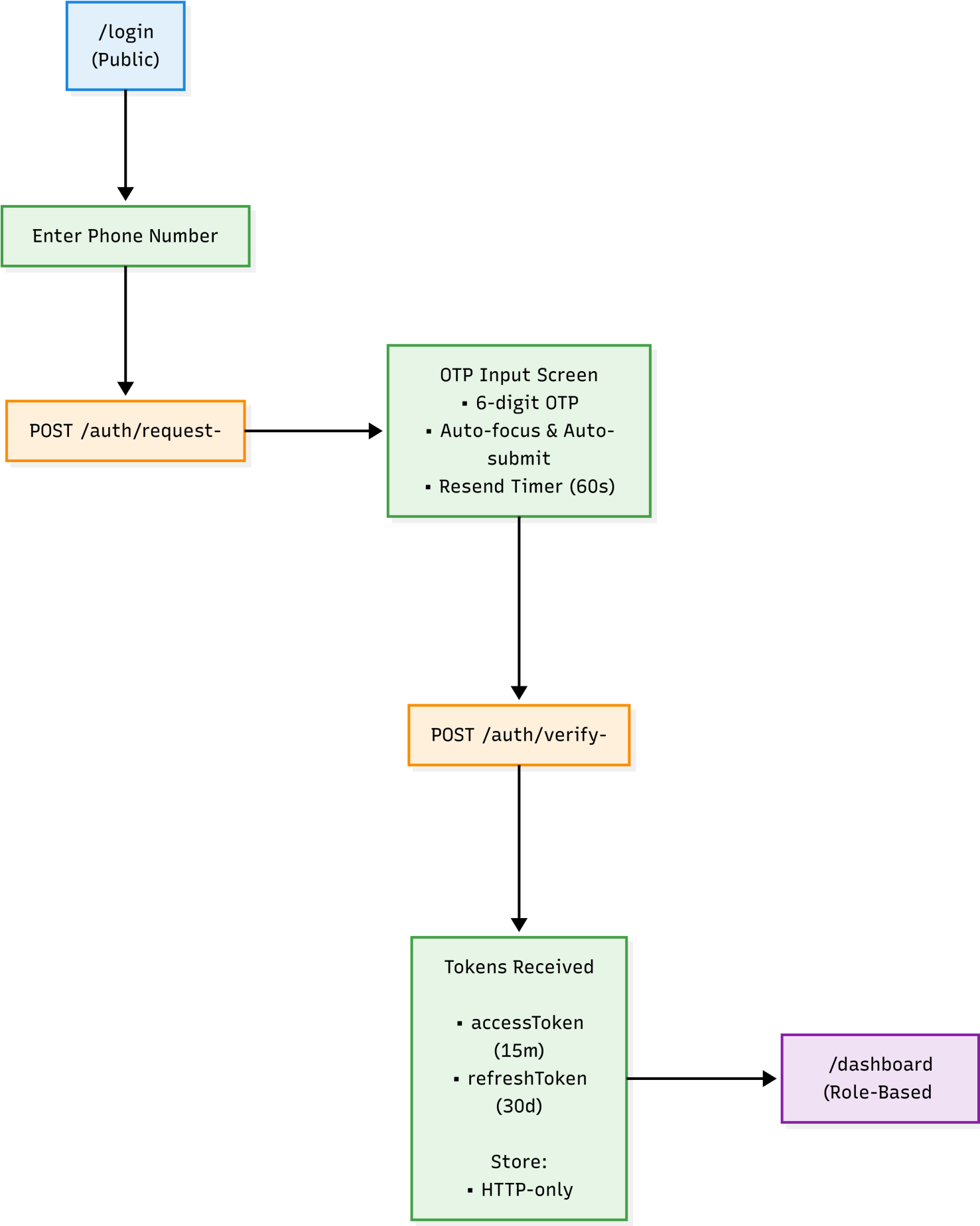
1. Application Structure

1.1 Total Pages/Screens

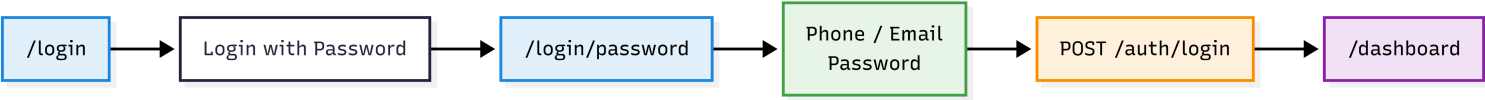
#	Page	Route	Type
1	Login (OTP)	/login	Public
2	Login (Password)	/login/password	Public
3	Register	/register	Public
4	Public IAS-P Assessment	/assess	Public
5	Dashboard	/dashboard	Protected
6	Patient List	/patients	Protected
7	Create Patient	/patients/new	Protected
8	Patient Detail	/patients/[id]	Protected
9	Create Visit	/patients/[id]/visits/new	Protected
10	Visit Detail	/patients/[id]/visits/[visitId]	Protected
11	Record Vitals	/patients/[id]/vitals/new	Protected
12	Submit IAS-P Assessment	/patients/[id]/scores/new	Protected
13	Score Detail	/patients/[id]/scores/[scoreId]	Protected
14	Task List	/tasks	Protected
15	Task Detail	/tasks/[id]	Protected
16	Alert List	/alerts	Protected
17	Alert Rules	/alerts/rules	Protected
18	Notifications	/notifications	Protected
19	Documents (per-patient)	/patients/[id]/documents	Protected
20	User Management	/admin/users	Protected (Admin)
21	Audit Logs	/admin/audit-logs	Protected (Admin)
22	Settings / Profile	/settings	Protected
23	404 Not Found	*	Public
24	403 Forbidden	— (component)	Protected

Total: 24 pages/screens

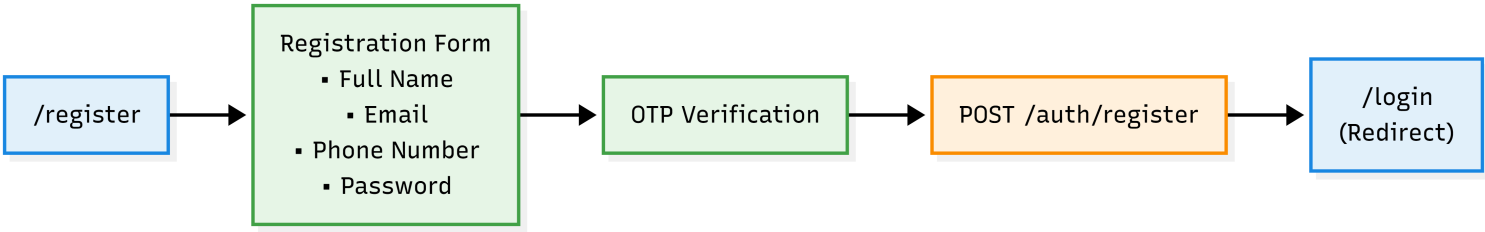
1.2 Authentication Flow



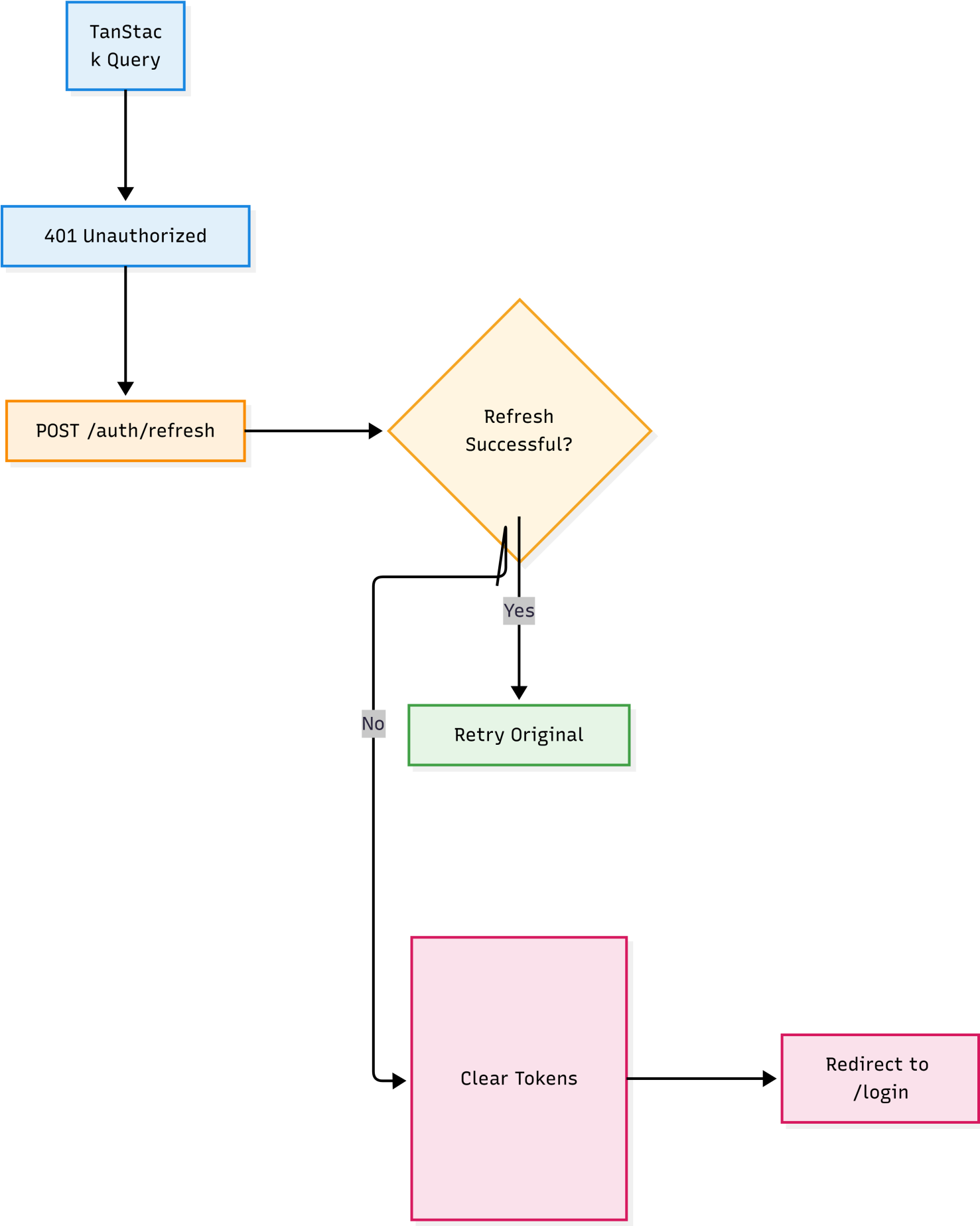
Password fallback flow:



Registration flow:



Token Refresh (background):



1.3 Public vs Protected Pages

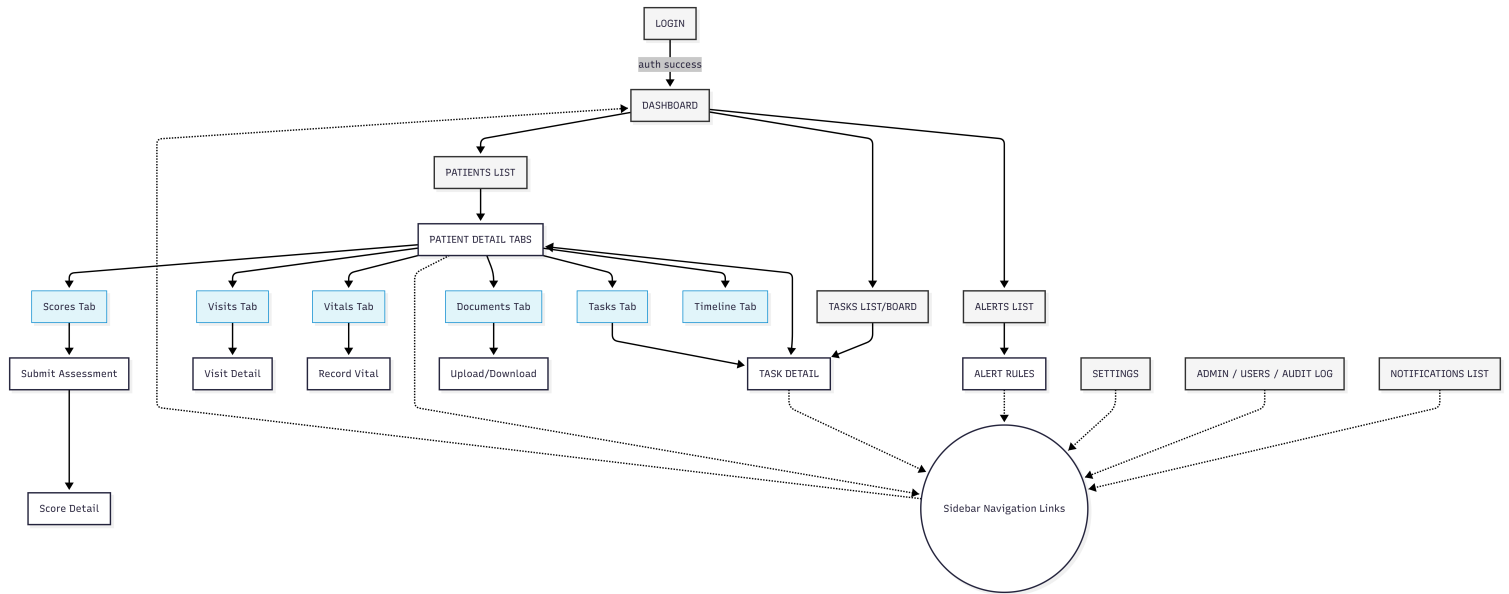
Type	Pages	Auth Required
Public	/login , /login/password , /register , /assess , /404	✗
Protected	All others (/dashboard , /patients/* , /tasks , /alerts , /notifications , /admin/* , /settings)	✓ JWT
Admin-only	/admin/users , /admin/audit-logs	✓ JWT + admin role
Coordinator+	/alerts/rules	✓ JWT + coordinator / admin

1.4 Route Hierarchy

/	→ Redirect to /dashboard (if auth) or /login
└─ /login	→ OTP Login [Public]

└─ /login/password	→ Password Login	[Public]
└─ /register	→ Registration	[Public]
└─ /assess	→ Public IAS-P Calculator	[Public]
└─ /dashboard	→ Role-based Dashboard	[Protected]
└─ /patients	→ Patient List	[Protected, nurse+]
└─ /patients/new	→ Create Patient	[Protected, nurse+]
└─ /patients/[id]	→ Patient Detail (tabbed)	[Protected, ABAC]
└─ ?tab=overview	→ Patient Overview tab	
└─ ?tab=visits	→ Visits tab	
└─ /patients/[id]/visits/new	→ Create Visit	
└─ /patients/[id]/visits/[visitId]	→ Visit Detail	
└─ ?tab=vitals	→ Vitals tab (charts + list)	
└─ /patients/[id]/vitals/new	→ Record Vital	
└─ ?tab=scores	→ IAS-P Scores tab	
└─ /patients/[id]/scores/new	→ Submit Assessment	
└─ /patients/[id]/scores/[scoreId]	→ Score Detail	
└─ ?tab=tasks	→ Patient Tasks tab	
└─ ?tab=alerts	→ Patient Alerts tab	
└─ ?tab=documents	→ Documents tab	
└─ /patients/[id]/documents	→ Upload/List	
└─ ?tab=timeline	→ Merged Timeline tab	
└─ /tasks	→ Global Task List/Kanban	[Protected]
└─ /tasks/[id]	→ Task Detail	[Protected]
└─ /alerts	→ Global Alert List	[Protected]
└─ /alerts/rules	→ Alert Rules Management	[Protected, coord+]
└─ /notifications	→ Notification History	[Protected]
└─ /admin	→ Admin Section	[Protected, admin]
└─ /admin/users	→ User Management	[Protected, admin]
└─ /admin/audit-logs	→ Audit Log Viewer	[Protected, admin]
└─ /settings	→ User Profile & Preferences	[Protected]
└─ /*	→ 404 Not Found	[Public]

1.5 Navigation Flow Between Pages

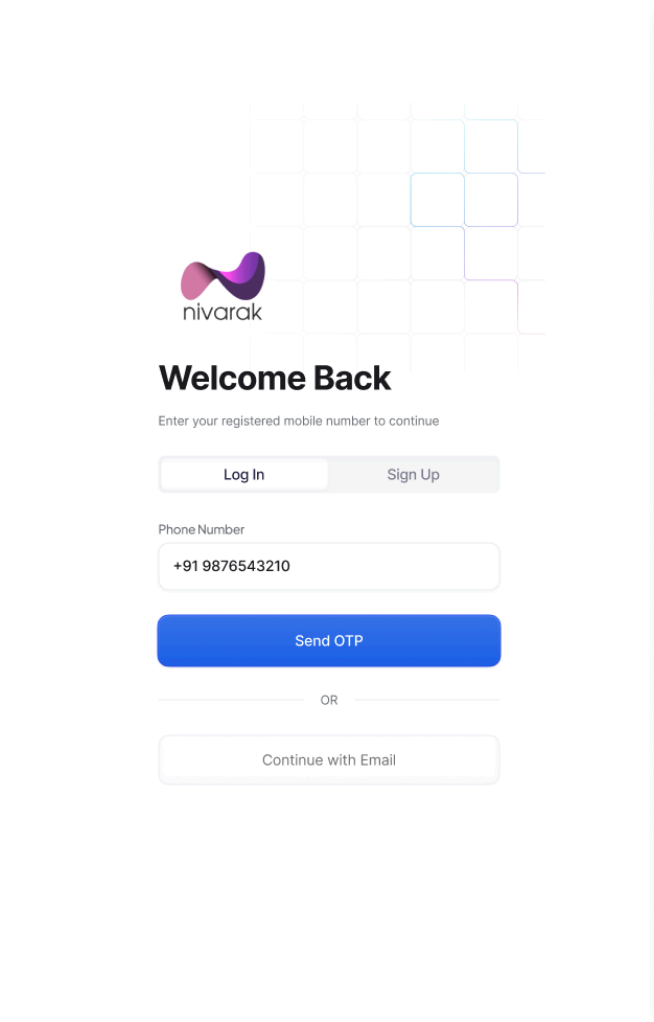


2. Page Mapping — Page-by-Page Breakdown

2.1 Login Page (OTP)

Attribute	Detail
Purpose	Primary authentication via phone OTP
Route	/login
Roles	Public (unauthenticated users)
Layout	Auth layout (centered card, no sidebar/header)

Wireframe-level layout:



API Calls:

API Method	When	Response	Usage
POST /auth/request-otp	User clicks "Send OTP"	Show OTP input, start timer	Initiates the one-time password generation and delivery.
POST /auth/verify-otp	User enters 6-digit OTP	Extract tokens, redirect to /dashboard	Validates the OTP and authenticates the user session.

State:

- Local: phone number, OTP digits[], isOtpSent, resendTimer, attempts
- Form: React Hook Form with Zod validation

Loading/Empty/Error States:

- Loading:** Button shows spinner during OTP send and verification
- Error:** Inline error below input (invalid phone, expired OTP, max attempts reached, rate limited)
- Empty:** N/A

2.2 Login Page (Password)

Attribute	Detail
Purpose	Alternative authentication via password
Route	/login/password
Roles	Public
Layout	Auth layout

API Calls: POST /auth/login (phone/email + password)

State: Local form state: phone, password, showPassword toggle

2.3 Register Page

Attribute	Detail
Purpose	New user account creation (post-OTP or post-invite)
Route	/register
Roles	Public
Layout	Auth layout

Wireframe:



Create Your Account

Create your account to continue!

Log In

Sign Up

Full Name

Lois Becket

Email

Loisbecket@gmail.com

Phone Number

+91 9876543210

Password

Confirm Password

☐ I agree to the [Terms & Conditions](#) and [Privacy Policy](#)

Create Account



API Calls: POST /auth/register

2.4 Public IAS-P Assessment Page

Attribute	Detail
Purpose	Free public IAS-P v2.0 calculator for awareness/lead-gen
Route	/assess
Roles	Public (anyone)
Layout	Public layout (minimal header, no sidebar)

Major UI Sections:

- 1. Introduction card explaining the assessment
- 2. Multi-step wizard form (same as authenticated version)
- 3. Results display with risk band visualization
- 4. CTA to register/login to save results

API Calls: POST /scores/calculate (dry-run, no auth)

State: Multi-step form state (24 params + proxy metadata + red flags), calculated result

2.5 Dashboard

Attribute	Detail
Purpose	Role-based overview with KPIs, recent activity, alerts
Route	/dashboard
Roles	All authenticated users (content varies by role)
Layout	App layout (sidebar + header)

Wireframe (Coordinator/Admin Dashboard):

Missing Wireframe Blueprint

Dashboard Variants:

Role	API Endpoint	KPIs Shown
Coordinator / Admin	GET /dashboard/coordinator	Total patients, risk band distribution, active alerts by severity, pending tasks, overdue tasks, recent assessments, care pathway distribution
Doctor / Nurse	GET /dashboard/clinician	My assigned patients, my open tasks, today's visits, vitals requiring attention, pending assessments
Caregiver	GET /dashboard/caregiver	Linked patients' status, latest vitals, latest scores, upcoming visits, alerts for my patients
Patient	GET /dashboard/caregiver (same)	My vitals, my scores, upcoming visits, my alerts

API Calls:

API	Method	Condition
GET /dashboard/coordinator	GET	Role is coordinator or admin
GET /dashboard/clinician	GET	Role is doctor or nurse
GET /dashboard/caregiver	GET	Role is caregiver or patient
GET /notifications	GET	All roles (for header badge count)

State:

- Server state: Dashboard data (TanStack Query, cached 30s, refetch on window focus)
- Global: User role, notification unread count

2.6 Patient List Page

Attribute	Detail
Purpose	Browse, search, and filter all patients
Route	/patients
Roles	nurse , doctor , coordinator , admin (RBAC level ≥ 2). Caregivers see only their linked patients
Layout	App layout

Wireframe:

Missing Wireframe Blueprint

API Calls:

API	Method	Params
GET /patients	GET	page , limit , search , carePathway , isActive

State:

- Server: Patient list (paginated, cached)
- Local: Search query, filter values, pagination state

Loading/Empty/Error:

- **Loading:** Skeleton table rows (8 rows)
- **Empty:** "No patients found" illustration + "Add your first patient" CTA
- **Error:** Error banner with retry button

2.7 Patient Detail Page (Tabbed)

Attribute	Detail
Purpose	Comprehensive patient view with all sub-data
Route	/patients/[id]
Roles	All authenticated (ABAC: caregivers only see linked patients)
Layout	App layout

Wireframe:

Missing Wireframe Blueprint

Tab API Calls:

Tab	APIs
Overview	GET /patients/:id , GET /patients/:id/vitals/latest , GET /patients/:id/scores/latest , GET /patients/:id/caregivers
Visits	GET /patients/:id/visits (paginated)
Vitals	GET /patients/:id/vitals (filtered), GET /patients/:id/vitals/trends
Scores	GET /patients/:id/scores (paginated)
Tasks	GET /tasks?patientId=:id
Alerts	GET /patients/:id/alerts
Documents	GET /patients/:id/documents
Timeline	GET /patients/:id/timeline

2.8 Create/Edit Patient Page

Attribute	Detail
Purpose	Register new patient or edit existing
Route	/patients/new or /patients/[id]/edit
Roles	nurse , doctor , coordinator , admin

Form Fields: full_name, date_of_birth, gender, blood_group, primary_language, phone (optional), address (JSONB: street, city, state, pincode), medical_history (JSONB: conditions[], allergies[], medications[]), emergency_contact (JSONB: name, phone, relationship)

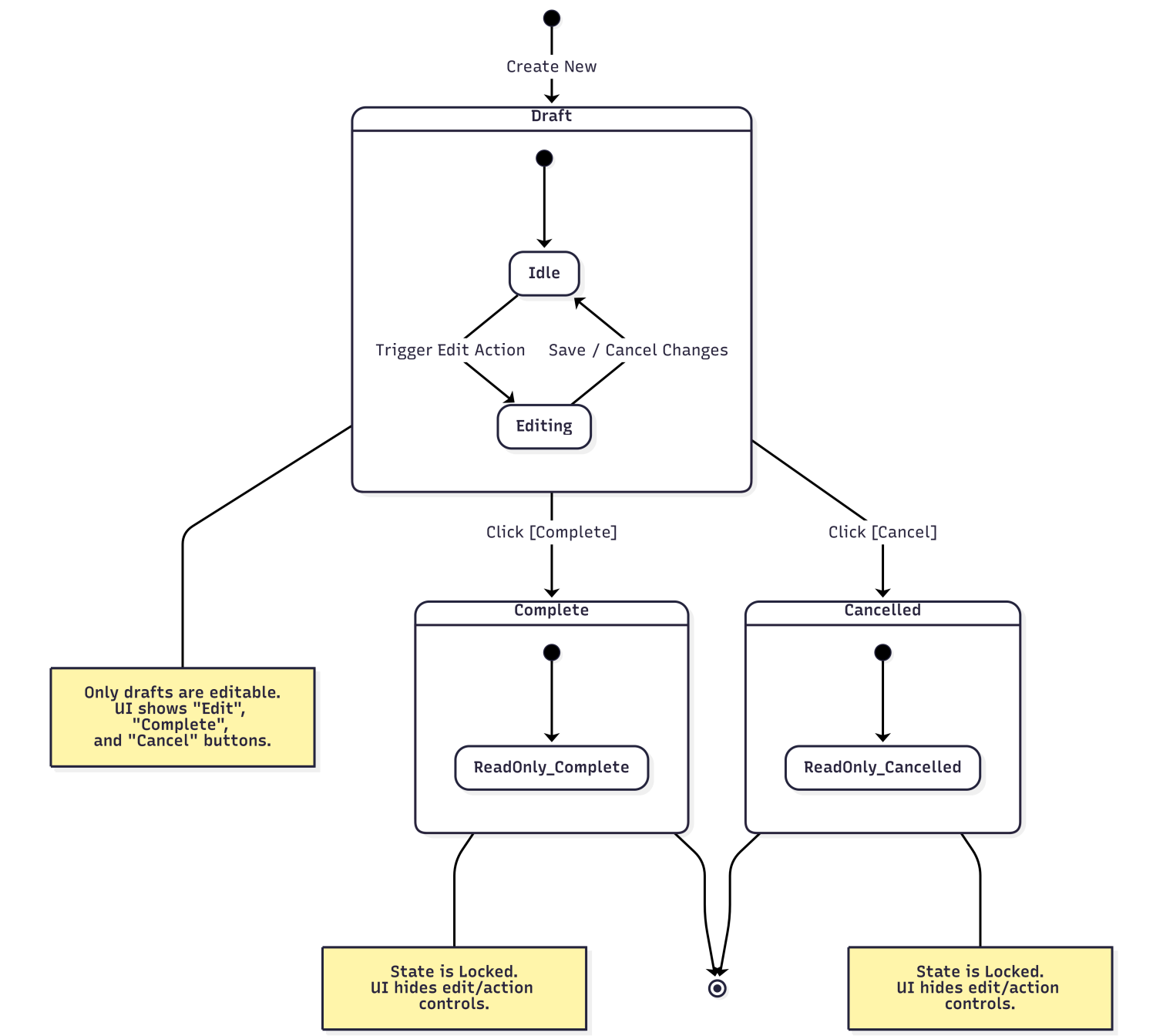
API: POST /patients or PUT /patients/:id

2.9 Visit Detail & Create Visit

Attribute	Detail
Route	/patients/[id]/visits/[visitId] and /patients/[id]/visits/new
Roles	nurse , doctor , coordinator , admin (ABAC)

Form Fields: visit_type (home_visit, clinic_visit, telehealth), visited_at, location, chief_complaint, clinical_notes, systemic_exam (JSONB), medications_reviewed (JSONB), attending_clinicians (JSONB array)

State Machine UI:



APIs:

POST /patients/:id/visits , PUT /patients/:id/visits/:visitId , POST /patients/:id/visits/:visitId/complete

2.10 Record Vitals Page

Attribute	Detail
Route	/patients/[id]/vitals/new
Roles	nurse , doctor , caregiver (ABAC)

Form Fields: parameter_type (dropdown of 8 types), value (numeric with physiological range validation), unit (auto-filled), visit_id (optional link), source (manual, device), device_id (optional)

Vital Types with Validation:

Type	Unit	Range
bp_systolic	mmHg	50-300
bp_diastolic	mmHg	20-200
heart_rate	bpm	20-250
spo2	%	50-100
temperature	°C	30-45
respiratory_rate	/min	5-60

Type	Unit	Range
weight	kg	10-300
blood_glucose	mg/dL	20-600

API: POST /patients/:id/vitals

2.11 Submit IAS-P Assessment (Multi-Step Wizard)

Attribute	Detail
Route	/patients/[id]/scores/new
Roles	caregiver, nurse, doctor, coordinator, admin (ABAC)

Wizard Steps:

Step 1: Proxy Metadata

- proxy_relationship (son/daughter/spouse/other)
- proxy_proximity (lives_with/nearby/remote)
- parent_age (numeric)
- living_situation (alone/with_spouse/with_family/care_facility)
- visit_frequency (daily/weekly/monthly/rarely)

Step 2: A – Basic Self-Care (4 items × 0–2 scale)

- bathing, dressing, toileting, feeding

Step 3: B – Daily Life Function (4 items)

- phone_communication, daily_home_tasks, simple_purchases, organizing_essentials

Step 4: C – Mobility (4 items)

- moving_inside_house, getting_up, walking_outside, falls_6_months

Step 5: D – Thinking & Decision (3 items)

- remembering_routine, understanding_instructions, safe_decisions

Step 6: E – Health Management (3 items)

- taking_medicines, understanding_health, following_appointments

Step 7: F – Nutrition & Continence (3 items)

- eating_drinking, weight_appetite, bladder_bowel

Step 8: G – Social Function (2 items)

- communicating_needs, social_contact

Step 9: H – Safety & Support (1 item)

- emergency_help

Step 10: Red Flags Checklist (8 boolean flags)

- two_or_more_falls, missed_incorrect_medicines, unsafe_decisions, significant_weight_loss, social_isolation, recurrent_hospital_admissions, caregiver_struggling, no_emergency_response

Step 11: Review & Submit

- Summary of all answers
- Live-calculated IAS% preview (client-side)
- Domain score breakdown (bar chart)
- Risk band indicator

- Red flag count and urgency level
- [Submit Assessment] button

UI for each assessment item:

Missing Wireframe Blueprint

APIs: POST /patients/:id/scores (submit), POST /scores/calculate (dry-run preview)

State: Multi-step form state managed by React Hook Form with per-step Zod schemas, step index, completed steps set

2.12 Score Detail Page

Attribute	Detail
Route	/patients/[id]/scores/[scoreId]
Roles	All authenticated (ABAC)

Sections:

- Score summary card (IAS%, risk band badge, care pathway, date)
- Domain scores bar chart (8 bars, colored by score)
- Individual item scores table
- Red flags list with urgency badge
- Proxy metadata (who assessed, relationship, proximity)
- Score comparison with previous assessment (if exists)

API: GET /patients/:id/scores/:scoreId

2.13 Task List Page

Attribute	Detail
Route	/tasks
Roles	nurse , doctor , coordinator , admin

Views (toggleable):

Table View:

Missing Wireframe Blueprint

Kanban View:

Missing Wireframe Blueprint

APIs: GET /tasks (with filters: status, priority, assignedTo, patientId, category)

2.14 Alert List Page

Attribute	Detail
Route	/alerts

Attribute	Detail
Roles	nurse , doctor , coordinator , admin

Wireframe:

Missing Wireframe Blueprint

APIs: GET /alerts , PATCH /alerts/:id/acknowledge , PATCH /alerts/:id/resolve

2.15 Alert Rules Page

Attribute	Detail
Route	/alerts/rules
Roles	coordinator , admin

APIs: GET /alerts/rules , POST /alerts/rules

2.16 Notifications Page

Attribute	Detail
Route	/notifications
Roles	All authenticated

APIs: GET /notifications , PATCH /notifications/:id/read

2.17 User Management (Admin)

Attribute	Detail
Route	/admin/users
Roles	admin only

Sections:

1. User table (name, phone, email, roles, last login, status)
2. Invite user dialog (phone, role assignment, invite via OTP)
3. Edit user roles

APIs: POST /auth/invite , plus user listing (assumption: GET /users endpoint exists or can be derived from /patients + user data)

 Note

Assumption: The HLD doesn't explicitly list a GET /users endpoint for admin user listing. This is assumed to exist or will need to be added to the backend.

2.18 Audit Logs (Admin)

Attribute	Detail
Route	/admin/audit-logs
Roles	admin only

Wireframe:

Missing Wireframe Blueprint

API: GET /audit-logs (paginated, filtered by actor_id, action, entity_type, date range)

2.19 Settings / Profile Page

Attribute	Detail
Route	/settings
Roles	All authenticated

Sections:

1. Profile info (name, email, phone — read-only phone)
2. Change password
3. Language preference
4. Theme toggle (light/dark)
5. Active sessions list with logout-all option
6. Notification preferences

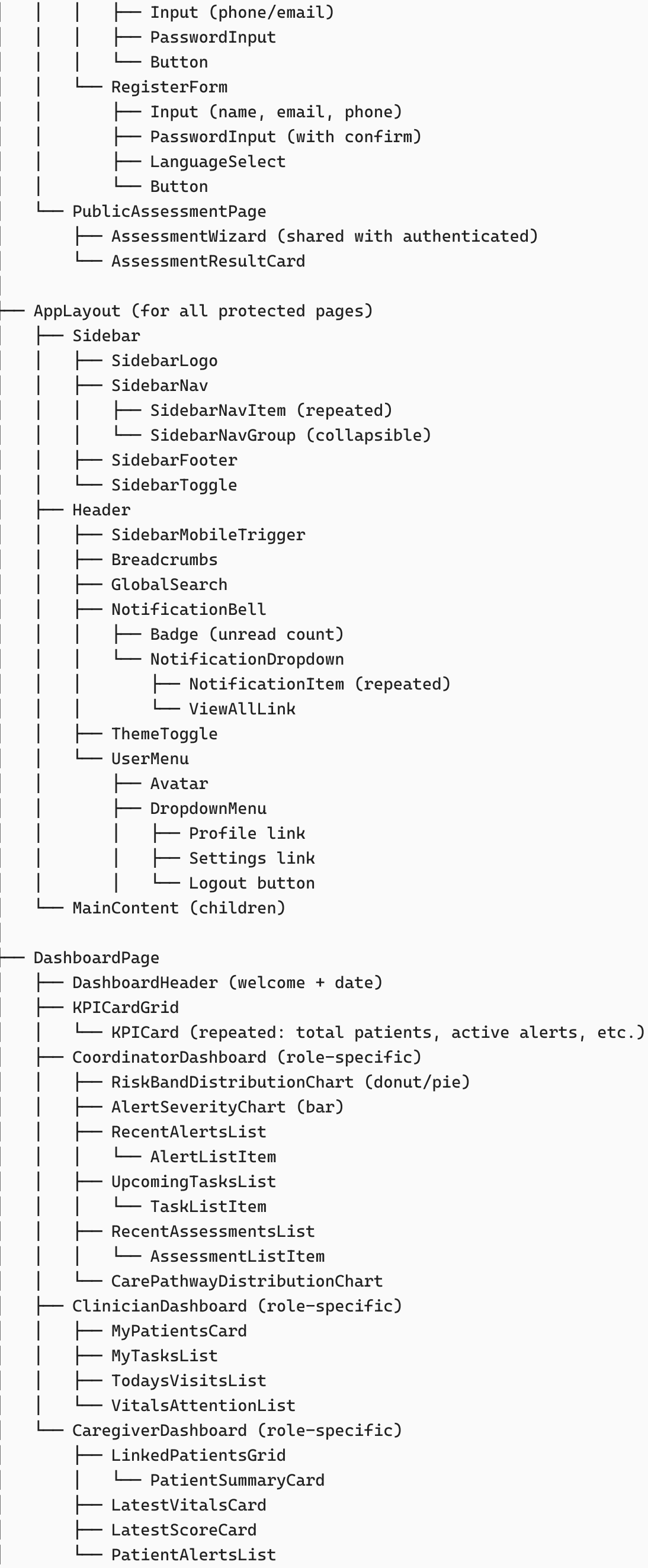
 **Note**

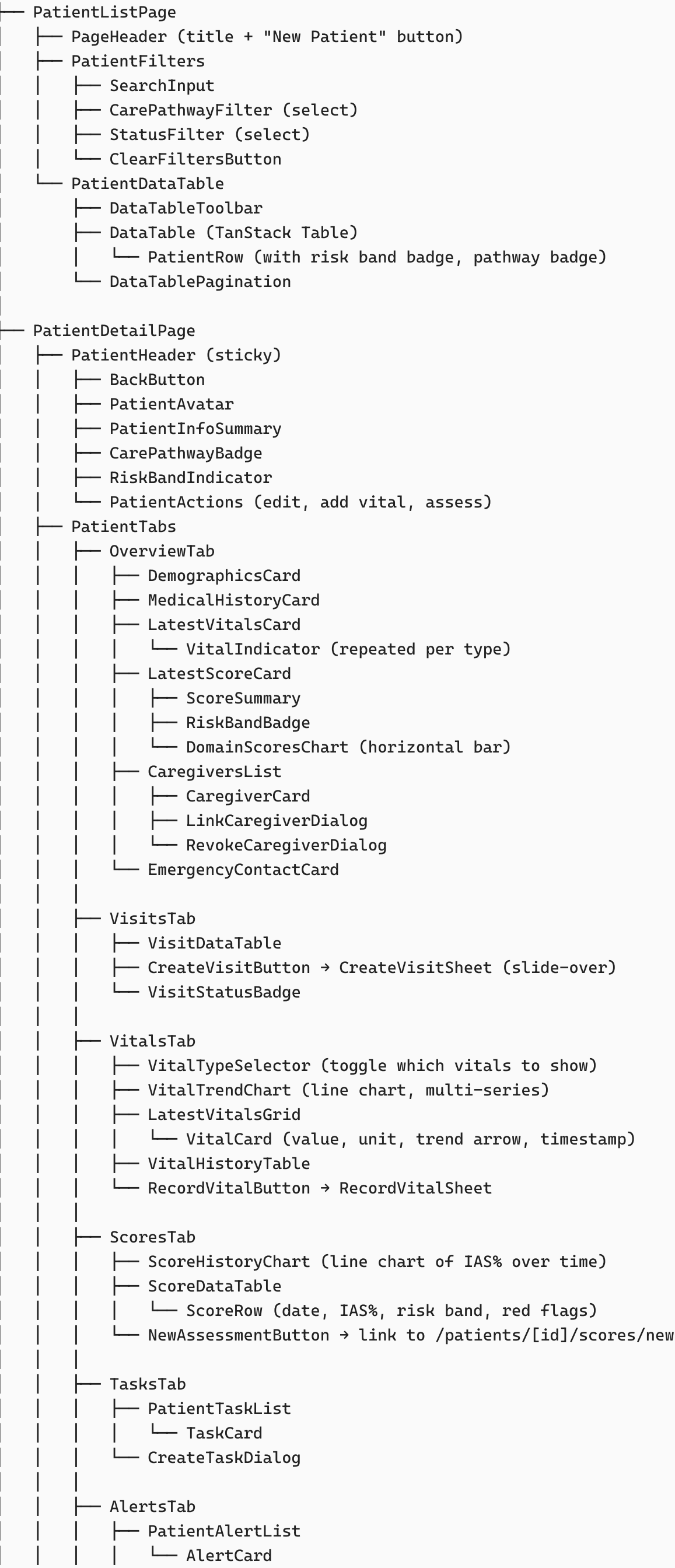
Assumption: A GET /auth/me or profile endpoint is assumed. The HLD doesn't explicitly list one, but the auth middleware loads user data on every request.

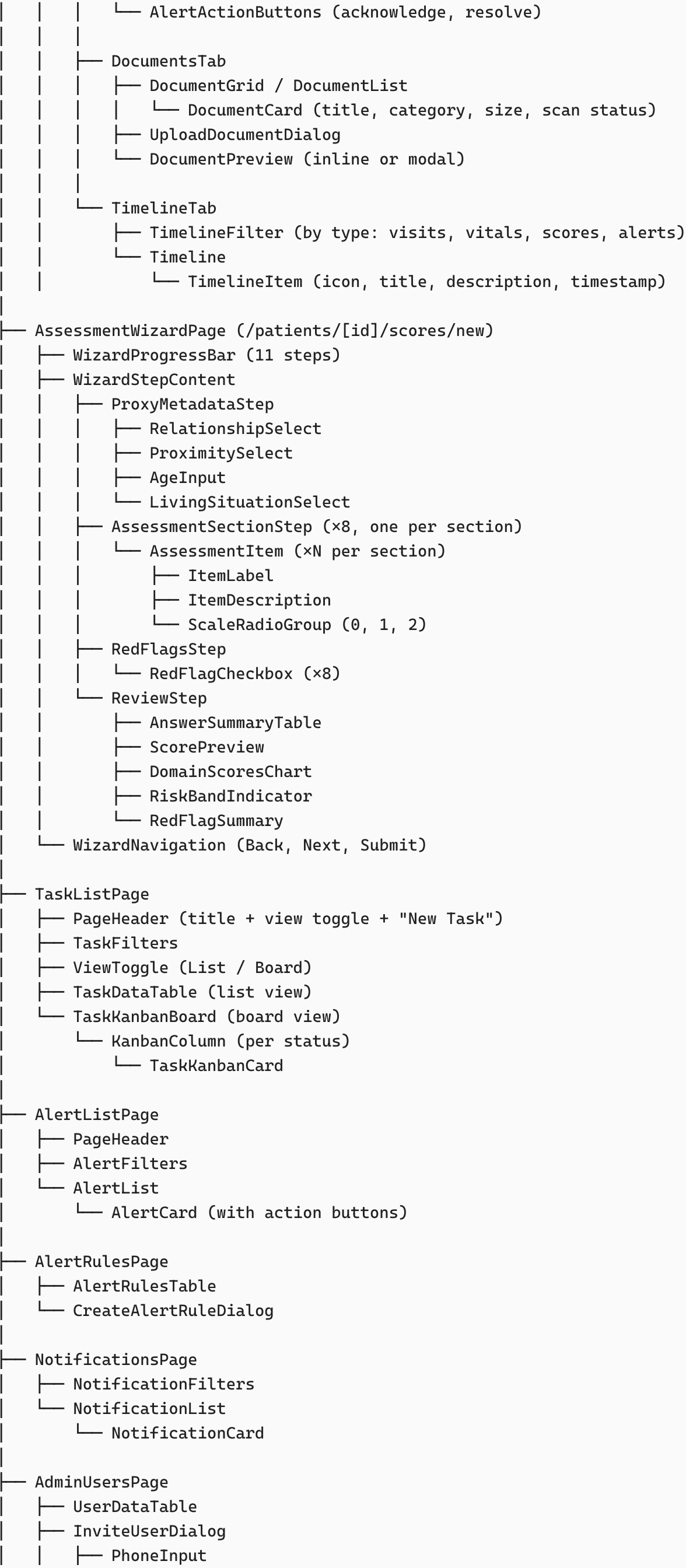
3. React Component Hierarchy

3.1 Component Tree (Parent-Child Relationships)

```
App (Next.js App Router)
├─ RootLayout
│   ├─ ThemeProvider (next-themes)
│   ├─ QueryClientProvider (TanStack Query)
│   └─ Toaster (sonner)
├─
└─ AuthLayout (for /login, /register, /assess)
    ├─ AuthCard
    │   ├─ Logo
    │   ├─ LoginOTPForm
    │   │   ├─ PhoneInput
    │   │   ├─ OTPInput (6-digit)
    │   │   └─ ResendTimer
    │   └─ Button
    └─ LoginPasswordForm
```







			RoleSelect
			Button
			EditUserRolesDialog
			AuditLogsPage
			AuditLogFilters
			ActorFilter
			ActionFilter
			EntityTypeFilter
			DateRangePicker
			AuditLogDataTable
			SettingsPage
			ProfileSection
			PasswordChangeSection
			PreferencesSection (language, theme)
			ActiveSessionsSection

4. Global (Shared) Components

Component	Purpose	Used On
Sidebar	Main navigation	All protected pages
SidebarNavItem	Individual nav link with icon + label	Sidebar
Header	Top bar with search, notifications, user menu	All protected pages
Breadcrumbs	Path trail navigation	All protected pages
GlobalSearch	Search patients, tasks across the app	Header
NotificationBell	Unread notification count + dropdown	Header
NotificationDropdown	Recent 10 notifications panel	Header
UserMenu	Avatar dropdown with profile/settings/logout	Header
ThemeToggle	Light/dark mode switch	Header / Settings
DataTable	Reusable server-paginated table	Patients, Tasks, Alerts, Audit, Visits, Scores, Documents
DataTableToolbar	Search + column visibility + filters	All DataTable instances
DataTablePagination	Page navigation + page size select	All DataTable instances
PageHeader	Page title + description + action buttons	All list pages
EmptyState	Illustration + message + CTA for empty lists	All list pages
ErrorBoundary	Catch React errors, show fallback UI	Wraps all pages
ErrorState	Error message + retry button	All data-fetching pages
LoadingSpinner	Centered spinner for page loads	Page transitions
SkeletonLoader	Skeleton placeholders during data load	Cards, tables, charts
ConfirmDialog	"Are you sure?" confirmation	Delete, revoke, cancel actions
Toasts	Success/error/info notifications (sonner)	After all mutations

Component	Purpose	Used On
Badge	Status/severity/role labels	Throughout
FilterBar	Horizontal filter controls	List pages
DateRangePicker	Date range selection	Audit logs, vital history
Avatar	User/patient avatar	Patient header, user menu, tables
KPICard	Stat card (value + label + trend)	Dashboard
StatusBadge	Colored status indicator	Tasks, visits, alerts
RiskBandBadge	Color-coded risk band label	Patient list, scores, dashboard
CarePathwayBadge	Care pathway label	Patient list, detail

5. Layout Mapping

5.1 Layout Types

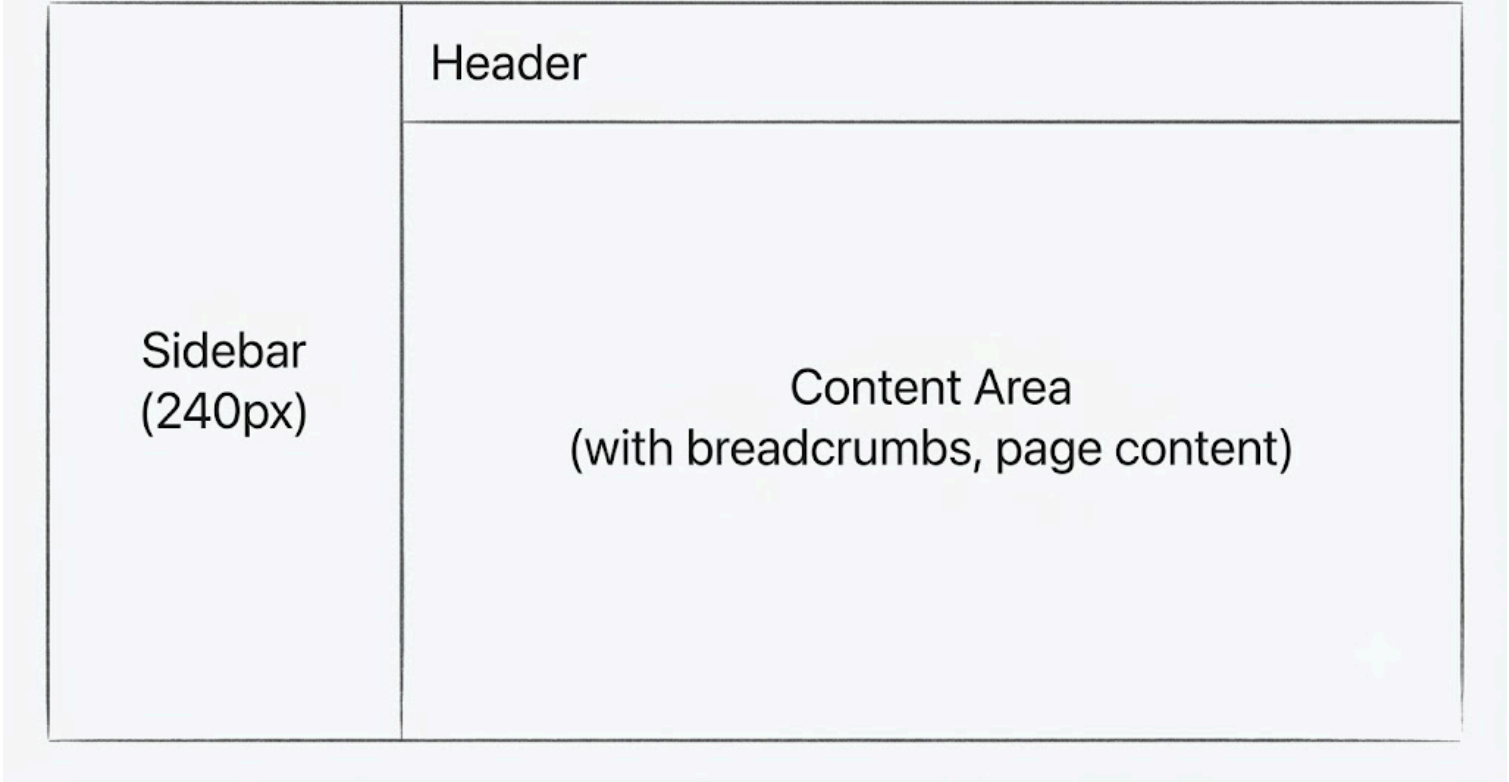
Layout	Pages	Sidebar	Header	Footer
AuthLayout	/login , /login/password , /register	✗	✗ (logo only)	✗
PublicLayout	/assess , /404	✗	✓ (minimal, with logo + login CTA)	✓
AppLayout	All protected pages	✓ (collapsible)	✓ (full)	✗

5.2 Responsive Behavior

Breakpoint	Sidebar	Header	Tables	Forms	Charts
Desktop (≥1280px)	Expanded (240px)	Full	Full table	Side-by-side fields	Full width
Tablet (768-1279px)	Collapsed (icons only, 64px)	Full, compact	Full table (horizontal scroll)	Stacked fields	Full width
Mobile (<768px)	Hidden (hamburger menu)	Compact (icon buttons)	Card list view	Full-width stacked	Full width, simplified

5.3 Mobile vs Desktop Layout

Desktop:



Mobile:



6. Feature Mapping

6.1 Authentication Feature

Aspect	Detail
Components	LoginOTPForm, LoginPasswordForm, RegisterForm, OTPInput, PhoneInput, ResendTimer
State	Zustand: useAuthStore (user, tokens, roles, isAuthenticated)
APIs	POST /auth/request-otp, POST /auth/verify-otp, POST /auth/login, POST /auth/register, POST /auth/refresh, POST /auth/logout
Navigation	Login → Dashboard (on success), Dashboard → Login (on token expiry)
Permissions	Public routes only
Validation	Phone: Indian format (+91), OTP: 6 digits, Password: min 8 chars
Dependencies	None (entry point)

6.2 Patient Management Feature

Aspect	Detail
Components	PatientDataTable, PatientFilters, CreatePatientForm, PatientHeader, DemographicsCard, MedicalHistoryCard, CaregiversList, LinkCaregiverDialog, RevokeCaregiverDialog
State	TanStack Query: <code>usePatients()</code> , <code>usePatient(id)</code> , <code>useCaregivers(patientId)</code> . Local: filters, search
APIs	GET/POST /patients, GET/PUT /patients/:id, GET/POST/DELETE /patients/:id/caregivers
Navigation	Patient List → Patient Detail → (Visits, Vitals, Scores, Tasks, Alerts, Documents)
Permissions	RBAC: nurse+ can CRUD. ABAC: caregiver/patient see only linked.
Validation	Required: full_name, date_of_birth, gender. Optional: medical_history (JSONB)
Dependencies	Auth (user context), used by Visits, Vitals, Scores, Tasks, Alerts, Documents

6.3 Visit Management Feature

Aspect	Detail
Components	VisitDataTable, CreateVisitSheet, VisitDetailView, VisitStatusBadge, VisitForm
State	TanStack Query: <code>useVisits(patientId)</code> , <code>useVisit(patientId, visitId)</code>
APIs	GET/POST /patients/:id/visits, GET/PUT /patients/:id/visits/:visitId, POST /patients/:id/visits/:visitId/complete
Navigation	Patient Detail (Visits tab) → Create Visit / Visit Detail
Permissions	ABAC: requires patient access
Validation	Required: visit_type, visited_at. State machine: only drafts editable
Dependencies	Patient (parent), Vitals (can be linked to visit)

6.4 Vitals Feature

Aspect	Detail
Components	VitalTrendChart, LatestVitalsGrid, VitalCard, VitalHistoryTable, RecordVitalSheet, VitalTypeSelector
State	TanStack Query: <code>useVitals(patientId)</code> , <code>useLatestVitals(patientId)</code> , <code>useVitalTrends(patientId)</code>
APIs	GET /patients/:id/vitals, GET /patients/:id/vitals/latest, GET /patients/:id/vitals/trends, POST /patients/:id/vitals
Navigation	Patient Detail (Vitals tab) → Record Vital (sheet)
Permissions	ABAC: requires patient access
Validation	Physiological range per vital type (e.g., bp_systolic: 50-300)
Dependencies	Patient (parent), Visit (optional link), triggers Alerts (via backend events)

6.5 IAS-P Scoring Feature

Aspect	Detail
Components	AssessmentWizard, WizardProgressBar, ProxyMetadataStep, AssessmentSectionStep, AssessmentItem, ScaleRadioGroup, RedFlagsStep, ReviewStep, ScorePreview, DomainScoresChart, RiskBandIndicator, ScoreHistoryChart, ScoreDetailView

Aspect	Detail
State	React Hook Form: multi-step form state (24 params + proxy + red flags). TanStack Query: <code>useScores(patientId)</code> , <code>useLatestScore(patientId)</code>
APIs	POST <code>/scores/calculate</code> (public dry-run), POST <code>/patients/:id/scores</code> , GET <code>/patients/:id/scores</code> , GET <code>/patients/:id/scores/latest</code> , GET <code>/patients/:id/scores/:scoreId</code>
Navigation	Patient Detail (Scores tab) → New Assessment → Review → Score Detail
Permissions	ABAC for patient-scoped. Public calculate endpoint has no auth
Validation	All 24 params required (0-2 scale), proxy fields required, red flags boolean
Dependencies	Patient (parent), triggers Alerts (red flags → backend events), updates care pathway

6.6 Task Management Feature

Aspect	Detail
Components	TaskDataTable, TaskFilters, TaskKanbanBoard, KanbanColumn, TaskKanbanCard, TaskCard, TaskDetailView, CreateTaskDialog, AssignTaskDialog
State	TanStack Query: <code>useTasks()</code> , <code>useTask(id)</code> . Local: view mode (list/board), filters
APIs	GET/POST <code>/tasks</code> , GET <code>/tasks/:id</code> , PATCH <code>/tasks/:id/assign</code> , PATCH <code>/tasks/:id/complete</code> , PATCH <code>/tasks/:id/escalate</code>
Navigation	Task List ↔ Task Detail, Patient Detail (Tasks tab) → Task Detail
Permissions	RBAC: nurse+ can create/view. Coordinator+ can assign.
Validation	Required: title, category, priority, patient_id
Dependencies	Patient, User (assignee), triggers Alerts/Notifications on overdue

6.7 Alerts Feature

Aspect	Detail
Components	AlertList, AlertCard, AlertFilters, AlertActionButtons, AlertRulesTable, CreateAlertRuleDialog
State	TanStack Query: <code>useAlerts()</code> , <code>usePatientAlerts(patientId)</code> , <code>useAlertRules()</code>
APIs	GET <code>/alerts</code> , GET <code>/patients/:id/alerts</code> , PATCH <code>/alerts/:id/acknowledge</code> , PATCH <code>/alerts/:id/resolve</code> , GET/POST <code>/alerts/rules</code>
Navigation	Alert List → Patient Detail (from alert's patient link)
Permissions	All authenticated can view alerts. Coordinator+ can manage alert rules
Validation	Alert rules: parameter_type, condition, threshold_value, severity required
Dependencies	Patient, Vitals (threshold breach), Scoring (red flags), Tasks (overdue)

6.8 Documents Feature

Aspect	Detail
Components	DocumentGrid, DocumentCard, UploadDocumentDialog, DocumentPreview
State	TanStack Query: <code>useDocuments(patientId)</code>
APIs	GET/POST <code>/patients/:id/documents</code> , GET <code>/patients/:id/documents/:docId/download</code>
Navigation	Patient Detail (Documents tab)
Permissions	ABAC: requires patient access
Validation	File size ≤ 50MB, allowed MIME types
Dependencies	Patient (parent), Visit (optional link)

6.9 Notification Feature

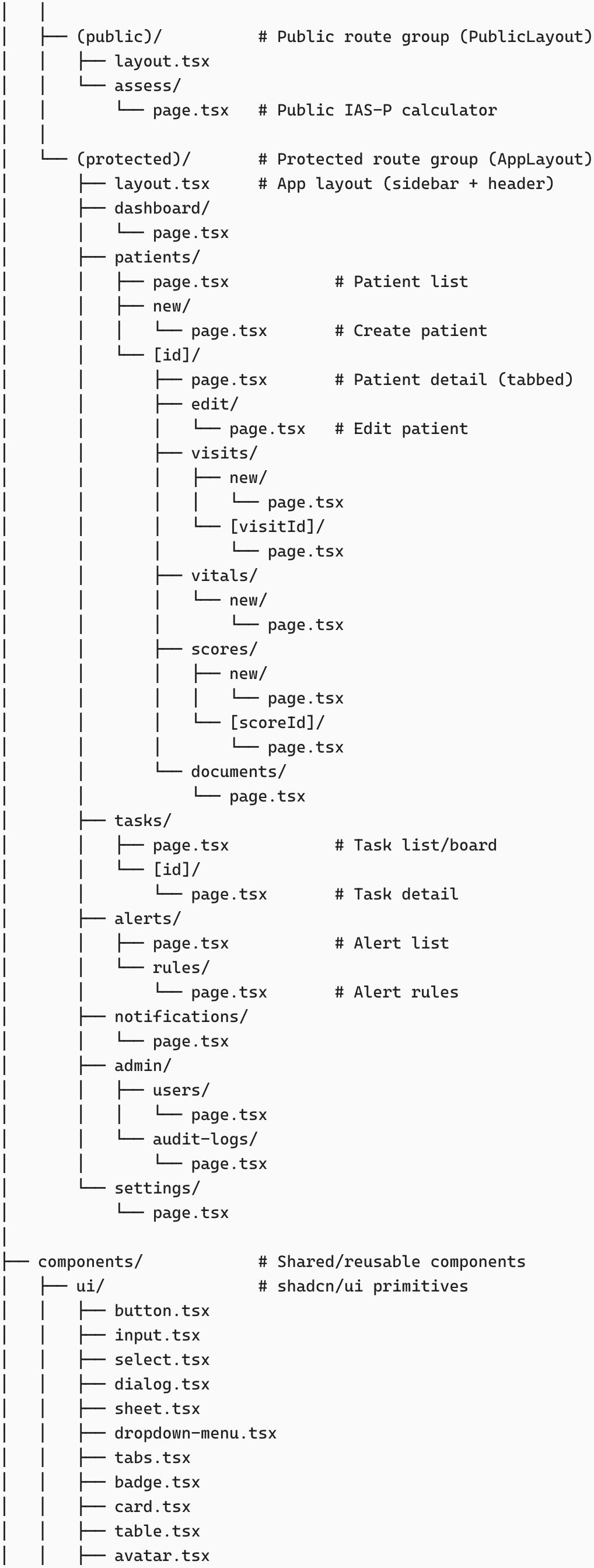
Aspect	Detail
Components	NotificationBell, NotificationDropdown, NotificationItem, NotificationList, NotificationFilters
State	TanStack Query: <code>useNotifications()</code> with <code>refetchInterval: 30000</code> . Zustand: <code>unreadCount</code>
APIs	GET <code>/notifications</code> , PATCH <code>/notifications/:id/read</code>
Navigation	Header bell → Dropdown → "View All" → <code>/notifications</code>
Permissions	Each user sees only their own notifications
Validation	N/A (read-only except mark-as-read)
Dependencies	Alerts (triggers notifications), Tasks (triggers notifications)

6.10 Admin Feature

Aspect	Detail
Components	UserDataTable, InviteUserDialog, EditUserRolesDialog, AuditLogDataTable, AuditLogFilters
State	TanStack Query: <code>useUsers()</code> , <code>useAuditLogs()</code>
APIs	POST <code>/auth/invite</code> , GET <code>/audit-logs</code>
Navigation	Admin Users → Invite Dialog. Audit Logs (standalone)
Permissions	Admin only
Validation	Invite: phone required, role ≤ inviter's role level
Dependencies	Auth (role hierarchy)

7. Folder Structure

nivarak/	
├─ src/	# Backend (existing)
├─ web/	# Frontend (NEW - Next.js App)
└─ public/	
└─ favicon.ico	
└─ logo.svg	
└─ images/	
└─ empty-states/	# Illustrations for empty states
└─ src/	
└─ app/	# Next.js App Router pages
└─ layout.tsx	# Root layout (providers, fonts)
└─ page.tsx	# / → redirect to /dashboard or /login
└─ not-found.tsx	# 404 page
└─ error.tsx	# Global error boundary
└─ (auth)/	# Auth route group (AuthLayout)
└─ layout.tsx	# Auth layout (centered card)
└─ login/	
└─ page.tsx	# OTP login
└─ password/	
└─ page.tsx	# Password login
└─ register/	
└─ page.tsx	# Registration



					└─ tooltip.tsx
					└─ checkbox.tsx
					└─ radio-group.tsx
					└─ separator.tsx
					└─ skeleton.tsx
					└─ scroll-area.tsx
					└─ popover.tsx
					└─ command.tsx
					└─ calendar.tsx
					└─ form.tsx
					└─ label.tsx
					└─ textarea.tsx
					└─ switch.tsx
					└─ accordion.tsx
					└─ progress.tsx
					└─ chart.tsx
				└─ layout/	# Layout components
				└─ sidebar.tsx	
				└─ sidebar-nav.tsx	
				└─ header.tsx	
				└─ breadcrumbs.tsx	
				└─ page-header.tsx	
				└─ page-container.tsx	
				└─ shared/	# Shared domain components
				└─ data-table.tsx	
				└─ data-table-toolbar.tsx	
				└─ data-table-pagination.tsx	
				└─ data-table-column-header.tsx	
				└─ data-table-faceted-filter.tsx	
				└─ global-search.tsx	
				└─ notification-bell.tsx	
				└─ notification-dropdown.tsx	
				└─ user-menu.tsx	
				└─ theme-toggle.tsx	
				└─ empty-state.tsx	
				└─ error-state.tsx	
				└─ loading-spinner.tsx	
				└─ confirm-dialog.tsx	
				└─ date-range-picker.tsx	
				└─ status-badge.tsx	
				└─ risk-band-badge.tsx	
				└─ care-pathway-badge.tsx	
				└─ priority-badge.tsx	
				└─ severity-badge.tsx	
				└─ kpi-card.tsx	
				└─ charts/	# Chart components
				└─ vital-trend-chart.tsx	
				└─ domain-scores-chart.tsx	
				└─ risk-band-distribution-chart.tsx	
				└─ alert-severity-chart.tsx	
				└─ score-history-chart.tsx	
				└─ care-pathway-chart.tsx	
				└─ features/	# Feature-specific components
				└─ auth/	
				└─ login-otp-form.tsx	
				└─ login-password-form.tsx	
				└─ register-form.tsx	
				└─ otp-input.tsx	
				└─ phone-input.tsx	
				└─ resend-timer.tsx	

```
└─ dashboard/
  └─ coordinator-dashboard.tsx
  └─ clinician-dashboard.tsx
  └─ caregiver-dashboard.tsx
  └─ dashboard-header.tsx
  └─ kpi-card-grid.tsx
  └─ recent-alerts-list.tsx
  └─ upcoming-tasks-list.tsx
  └─ recent-assessments-list.tsx

└─ patients/
  └─ patient-data-table.tsx
  └─ patient-columns.tsx          # TanStack Table column defs
  └─ patient-filters.tsx
  └─ patient-header.tsx
  └─ patient-tabs.tsx
  └─ create-patient-form.tsx
  └─ demographics-card.tsx
  └─ medical-history-card.tsx
  └─ emergency-contact-card.tsx
  └─ caregiver-list.tsx
  └─ caregiver-card.tsx
  └─ link-caregiver-dialog.tsx
  └─ revoke-caregiver-dialog.tsx
  └─ patient-summary-card.tsx    # Used in caregiver dashboard

└─ visits/
  └─ visit-data-table.tsx
  └─ visit-columns.tsx
  └─ create-visit-sheet.tsx
  └─ visit-detail-view.tsx
  └─ visit-form.tsx
  └─ visit-status-badge.tsx

└─ vitals/
  └─ vital-card.tsx
  └─ latest-vitals-grid.tsx
  └─ vital-history-table.tsx
  └─ vital-type-selector.tsx
  └─ record-vital-sheet.tsx
  └─ vital-indicator.tsx

└─ scoring/
  └─ assessment-wizard.tsx
  └─ wizard-progress-bar.tsx
  └─ proxy-metadata-step.tsx
  └─ assessment-section-step.tsx
  └─ assessment-item.tsx
  └─ scale-radio-group.tsx
  └─ red-flags-step.tsx
  └─ review-step.tsx
  └─ score-preview.tsx
  └─ score-detail-view.tsx
  └─ score-data-table.tsx
  └─ score-columns.tsx
  └─ assessment-result-card.tsx

└─ tasks/
  └─ task-data-table.tsx
  └─ task-columns.tsx
  └─ task-filters.tsx
  └─ task-kanban-board.tsx
  └─ kanban-column.tsx
```

					└─ task-kanban-card.tsx
					└─ task-card.tsx
					└─ task-detail-view.tsx
					└─ create-task-dialog.tsx
					└─ assign-task-dialog.tsx
					└─ view-toggle.tsx
				└─ alerts/	
					└─ alert-list.tsx
					└─ alert-card.tsx
					└─ alert-filters.tsx
					└─ alert-action-buttons.tsx
					└─ alert-rules-table.tsx
					└─ create-alert-rule-dialog.tsx
				└─ documents/	
					└─ document-grid.tsx
					└─ document-card.tsx
					└─ upload-document-dialog.tsx
					└─ document-preview.tsx
				└─ notifications/	
					└─ notification-list.tsx
					└─ notification-card.tsx
					└─ notification-filters.tsx
				└─ admin/	
					└─ user-data-table.tsx
					└─ user-columns.tsx
					└─ invite-user-dialog.tsx
					└─ edit-user-roles-dialog.tsx
					└─ audit-log-data-table.tsx
					└─ audit-log-columns.tsx
					└─ audit-log-filters.tsx
				└─ settings/	
					└─ profile-section.tsx
					└─ password-change-section.tsx
					└─ preferences-section.tsx
					└─ active-sessions-section.tsx
				└─ timeline/	
					└─ timeline.tsx
					└─ timeline-item.tsx
					└─ timeline-filter.tsx
			└─ hooks/	# Custom React hooks	
				└─ use-auth.ts	# Auth state + login/logout actions
				└─ use-user.ts	# Current user info + roles
				└─ use-patients.ts	# Patient CRUD queries/mutations
				└─ use-patient.ts	# Single patient query
				└─ use-visits.ts	# Visit queries/mutations
				└─ use-vitals.ts	# Vital queries/mutations
				└─ use-scores.ts	# Score queries/mutations
				└─ use-tasks.ts	# Task queries/mutations
				└─ use-alerts.ts	# Alert queries/mutations
				└─ use-documents.ts	# Document queries/mutations
				└─ use-notifications.ts	# Notification queries + polling
				└─ use-dashboard.ts	# Dashboard data queries
				└─ use-audit-logs.ts	# Audit log queries
				└─ use-users.ts	# Admin user management
				└─ use-debounce.ts	# Debounced value hook
				└─ use-media-query.ts	# Responsive breakpoint hook
				└─ use-sidebar.ts	# Sidebar collapse state

	└─ services/	# API client layer
	└─ api-client.ts	# Axios/fetch wrapper with auth interceptor
	└─ auth.service.ts	
	└─ patient.service.ts	
	└─ visit.service.ts	
	└─ vital.service.ts	
	└─ score.service.ts	
	└─ task.service.ts	
	└─ alert.service.ts	
	└─ document.service.ts	
	└─ notification.service.ts	
	└─ dashboard.service.ts	
	└─ audit.service.ts	
	└─ user.service.ts	
	└─ stores/	# Zustand stores
	└─ auth-store.ts	# Auth tokens, user, isAuthenticated
	└─ ui-store.ts	# Sidebar state, theme, preferences
	└─ notification-store.ts	# Unread count badge
	└─ lib/	# Utility functions
	└─ utils.ts	# cn() classname merger, formatters
	└─ constants.ts	# Role hierarchy, vital types, etc.
	└─ validations.ts	# Shared Zod schemas
	└─ permissions.ts	# Role-based permission checks
	└─ formatters.ts	# Date, number, phone formatters
	└─ query-keys.ts	# TanStack Query key factories
	└─ types/	# TypeScript type definitions
	└─ api.ts	# API response envelope types
	└─ auth.ts	# User, Role, Session types
	└─ patient.ts	# Patient, CaregiverLink types
	└─ visit.ts	# Visit types
	└─ vital.ts	# Vital, VitalType types
	└─ score.ts	# AgingScore, DomainScore, RedFlag types
	└─ task.ts	# Task types
	└─ alert.ts	# Alert, AlertRule types
	└─ document.ts	# Document types
	└─ notification.ts	# Notification types
	└─ audit.ts	# AuditLog types
	└─ dashboard.ts	# Dashboard KPI types
	└─ styles/	
	└─ globals.css	# Tailwind directives + CSS custom properties
	└─ .env.local	# NEXT_PUBLIC_API_URL
	└─ next.config.ts	
	└─ tailwind.config.ts	
	└─ tsconfig.json	
	└─ postcss.config.js	
	└─ package.json	
	└─ components.json	# shadcn/ui config

8. Reusable Components Inventory

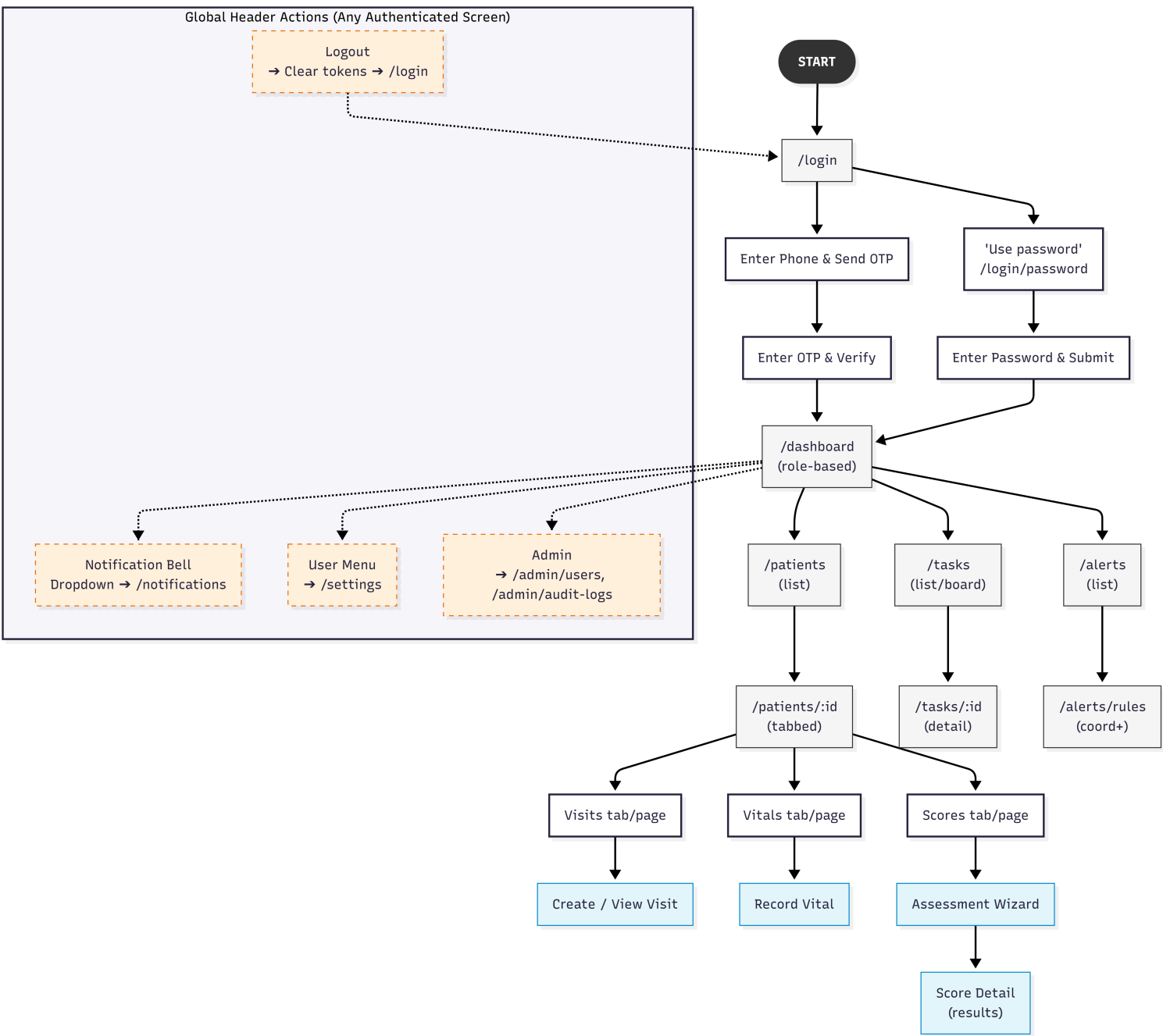
#	Component	Purpose	Key Props	Reusability	Pages Used
1	DataTable	Server-paginated data table	columns , data , pageCount , onPageChange	High	Patients, Tasks, Alerts, Visits, Scores, Documents, Audit Logs, Users
2	KPICard	Stat card with value + trend	title , value , trend , icon , variant	High	Dashboard (all variants)
3	StatusBadge	Colored status label	status , variant	High	Tasks, Visits, Alerts, Documents
4	RiskBandBadge	Risk band with color	band (strong_independent → high_dependence)	Medium	Patient list, detail, scores, dashboard
5	CarePathwayBadge	Care pathway label	pathway (home_care → high_dependency)	Medium	Patient list, detail, scores
6	PriorityBadge	Priority indicator	priority (low, medium, high, critical)	Medium	Tasks, alerts
7	SeverityBadge	Alert severity label	severity (info, warning, critical)	Medium	Alerts, notifications
8	EmptyState	No data illustration + CTA	title , description , action , icon	High	All list pages
9	ErrorState	Error message + retry	message , onRetry	High	All data-fetching pages
10	ConfirmDialog	Confirmation modal	title , description , onConfirm , variant	High	Delete, revoke, cancel actions
11	PageHeader	Page title + actions	title , description , children (action buttons)	High	All list/detail pages
12	LoadingSpinner	Centered spinner	size , className	High	Page transitions
13	SkeletonLoader	Content placeholder	variant (card, table-row, chart)	High	All loading states
14	DateRangePicker	Date range selector	from , to , onChange	Medium	Audit logs, vitals, visits
15	GlobalSearch	Search across entities	placeholder , onSearch	Low	Header
16	NotificationBell	Bell icon + unread count	count	Low	Header
17	UserMenu	Avatar + dropdown	user	Low	Header
18	ThemeToggle	Light/dark switch	—	Low	Header, settings
19	PhoneInput	Indian phone input	value , onChange , error	Medium	Login, register, invite, patient
20	OTPInput	6-digit OTP entry	value , onChange , length	Low	Login

#	Component	Purpose	Key Props	Reusability	Pages Used
21	VitalCard	Single vital reading	type , value , unit , timestamp , trend	Medium	Patient vitals, dashboard
22	AssessmentItem	IAS-P item with radio scale	label , description , value , onChange	Medium	Assessment wizard, public assess
23	WizardProgressBar	Multi-step progress	steps , currentStep , completedSteps	Medium	Assessment wizard
24	Timeline	Vertical timeline	items	Low	Patient timeline tab
25	TimelineItem	Single timeline entry	icon , title , description , timestamp , type	Low	Timeline

9. Page-to-Component Matrix

Component ↓ \ Page →	Login	Register	Public Assess	Dashboard	Patient List	Patient Detail	Visit Detail	Record Vital
AuthLayout	✓	✓						
AppLayout				✓	✓	✓	✓	✓
Sidebar				✓	✓	✓	✓	✓
Header				✓	✓	✓	✓	✓
Breadcrumbs				✓	✓	✓	✓	✓
DataTable					✓	✓		
KPICard				✓				
StatusBadge				✓		✓	✓	
RiskBandBadge			✓	✓	✓	✓		
EmptyState					✓	✓		
ConfirmDialog						✓		
PageHeader					✓			
NotificationBell				✓	✓	✓	✓	✓
DateRangePicker						✓		
PhoneInput	✓	✓				✓		
VitalCard				✓		✓		
AssessmentItem			✓					
Charts			✓	✓		✓		
WizardProgressBar			✓					

10. User Flow (Login to Logout)



11. State Management

11.1 State Categories

Category	Tool	What's Stored	Persistence
Auth State	Zustand (useAuthStore)	accessToken, refreshToken, user object, roles[], isAuthenticated	localStorage (tokens), memory (user)
UI State	Zustand (useUIStore)	sidebarCollapsed, theme (light/dark), language	localStorage
Notification Badge	Zustand (useNotificationStore)	unreadCount	Memory (refreshed from API)
Server State	TanStack Query	All API data (patients, visits, vitals, scores, tasks, alerts, notifications, dashboard, audit logs)	In-memory cache with configurable staleTime
Form State	React Hook Form	Form field values, validation errors, dirty/touched state	Component-local (assessment wizard may use sessionStorage for draft)
URL State	Next.js router + searchParams	Active tab, pagination page, sort column, filters	URL search parameters

11.2 TanStack Query Configuration

```
// Query key factories (in lib/query-keys.ts)
export const queryKeys = {
  patients: {
    all: ['patients'] as const,
    lists: () => [...queryKeys.patients.all, 'list'] as const,
    list: (filters: PatientFilters) => [...queryKeys.patients.lists(), filters] as const,
    details: () => [...queryKeys.patients.all, 'detail'] as const,
    detail: (id: string) => [...queryKeys.patients.details(), id] as const,
    caregivers: (id: string) => [...queryKeys.patients.detail(id), 'caregivers'] as const,
    timeline: (id: string) => [...queryKeys.patients.detail(id), 'timeline'] as const,
  },
  vitals: {
    byPatient: (patientId: string) => ['vitals', patientId] as const,
    latest: (patientId: string) => ['vitals', patientId, 'latest'] as const,
    trends: (patientId: string, params?: VitalTrendParams) => ['vitals', patientId, 'trends', params] as const,
  },
  scores: {
    byPatient: (patientId: string) => ['scores', patientId] as const,
    latest: (patientId: string) => ['scores', patientId, 'latest'] as const,
    detail: (patientId: string, scoreId: string) => ['scores', patientId, scoreId] as const,
  },
  tasks: {
    all: ['tasks'] as const,
    list: (filters: TaskFilters) => ['tasks', 'list', filters] as const,
    detail: (id: string) => ['tasks', id] as const,
  },
  alerts: {
    all: ['alerts'] as const,
    list: (filters: AlertFilters) => ['alerts', 'list', filters] as const,
    byPatient: (patientId: string) => ['alerts', 'patient', patientId] as const,
    rules: ['alerts', 'rules'] as const,
  },
  notifications: {
    all: ['notifications'] as const,
    list: (filters?: NotificationFilters) => ['notifications', 'list', filters] as const,
    unreadCount: ['notifications', 'unread-count'] as const,
  },
  dashboard: {
    coordinator: ['dashboard', 'coordinator'] as const,
    clinician: ['dashboard', 'clinician'] as const,
    caregiver: ['dashboard', 'caregiver'] as const,
  },
  auditLogs: {
    list: (filters: AuditLogFilters) => ['audit-logs', filters] as const,
  },
} as const;
```

11.3 Cache Strategy

Data	staleTime	gcTime	refetchOnWindowFocus	refetchInterval
Patient list	30s	5m	✓	—
Patient detail	60s	10m	✓	—
Vitals (latest)	30s	5m	✓	—
Vital trends	60s	10m	✓	—

Data	staleTime	gcTime	refetchOnWindowFocus	refetchInterval
Scores	60s	10m	✔	—
Tasks	30s	5m	✔	—
Alerts	15s	5m	✔	30s (critical)
Notifications	10s	5m	✔	30s
Dashboard	30s	5m	✔	60s
Audit logs	60s	10m	✔	—

12. API Integration

12.1 API Client Architecture

```
// services/api-client.ts
- Base URL from NEXT_PUBLIC_API_URL
- Axios instance with interceptors:
  1. Request interceptor: Attach Bearer token from auth store
  2. Response interceptor:
    - On 401: Attempt token refresh (POST /auth/refresh)
    - On refresh success: Retry original request
    - On refresh failure: Clear auth store, redirect to /login
- Standard response envelope: { success, data, error }
- Error transformation: Map API error codes to user-friendly messages
```

12.2 Page-to-API Mapping

Page	APIs	Method	Loading Strategy	Error Handling
Login (OTP)	/auth/request-otp , /auth/verify-otp	POST, POST	Button loading state	Inline error (toast for rate limit)
Login (Password)	/auth/login	POST	Button loading	Inline error
Register	/auth/register	POST	Button loading	Inline field errors
Public Assess	/scores/calculate	POST	Button loading + skeleton results	Error state card
Dashboard	/dashboard/{role} , /notifications	GET, GET	Skeleton KPI cards + list skeletons	Error boundary per section
Patient List	/patients	GET	Skeleton table rows	Error state with retry
Patient Detail	/patients/:id , /patients/:id/vitals/latest , /patients/:id/scores/latest , /patients/:id/caregivers	GET ×4	Parallel queries, skeleton per section	Per-section error states
Visits Tab	/patients/:id/visits	GET	Skeleton table	Error state
Vitals Tab	/patients/:id/vitals , /patients/:id/vitals/trends	GET ×2	Skeleton chart + table	Error state
Scores Tab	/patients/:id/scores	GET	Skeleton table	Error state

Page	APIs	Method	Loading Strategy	Error Handling
Create Visit	/patients/:id/visits	POST	Button loading	Toast error + field errors
Record Vital	/patients/:id/vitals	POST	Button loading	Toast error + field errors
Assessment	/patients/:id/scores , /scores/calculate	POST, POST	Step navigation loading, submit loading	Per-step validation, toast on submit error
Score Detail	/patients/:id/scores/:scoreId	GET	Skeleton card + chart	Error state
Task List	/tasks	GET	Skeleton table/kanban	Error state
Task Detail	/tasks/:id	GET	Skeleton card	Error state
Task Actions	/tasks/:id/assign , /tasks/:id/complete , /tasks/:id/escalate	PATCH	Button loading	Toast
Alert List	/alerts	GET	Skeleton list	Error state
Alert Actions	/alerts/:id/acknowledge , /alerts/:id/resolve	PATCH	Button loading	Toast
Alert Rules	/alerts/rules	GET, POST	Skeleton table, dialog loading	Error state, dialog errors
Notifications	/notifications , /notifications/:id/read	GET, PATCH	Skeleton list, inline loading	Error state
Admin Users	(user listing), /auth/invite	GET, POST	Skeleton table, dialog loading	Error state, dialog errors
Audit Logs	/audit-logs	GET	Skeleton table	Error state

13. Design System

13.1 Color Palette (Not Final)

Token	Light	Dark	Usage
--primary	#2563eb (Blue 600)	#3b82f6 (Blue 500)	Primary actions, links, active states
--primary-foreground	#ffffff	#ffffff	Text on primary
--destructive	#dc2626 (Red 600)	#ef4444	Danger actions, critical alerts
--warning	#d97706 (Amber 600)	#f59e0b	Warning alerts, overdue tasks
--success	#059669 (Emerald 600)	#10b981	Success states, completed
--muted	#64748b (Slate 500)	#94a3b8	Secondary text
--accent	#7c3aed (Violet 600)	#8b5cf6	IAS-P scores, special badges

13.2 Risk Band Colors

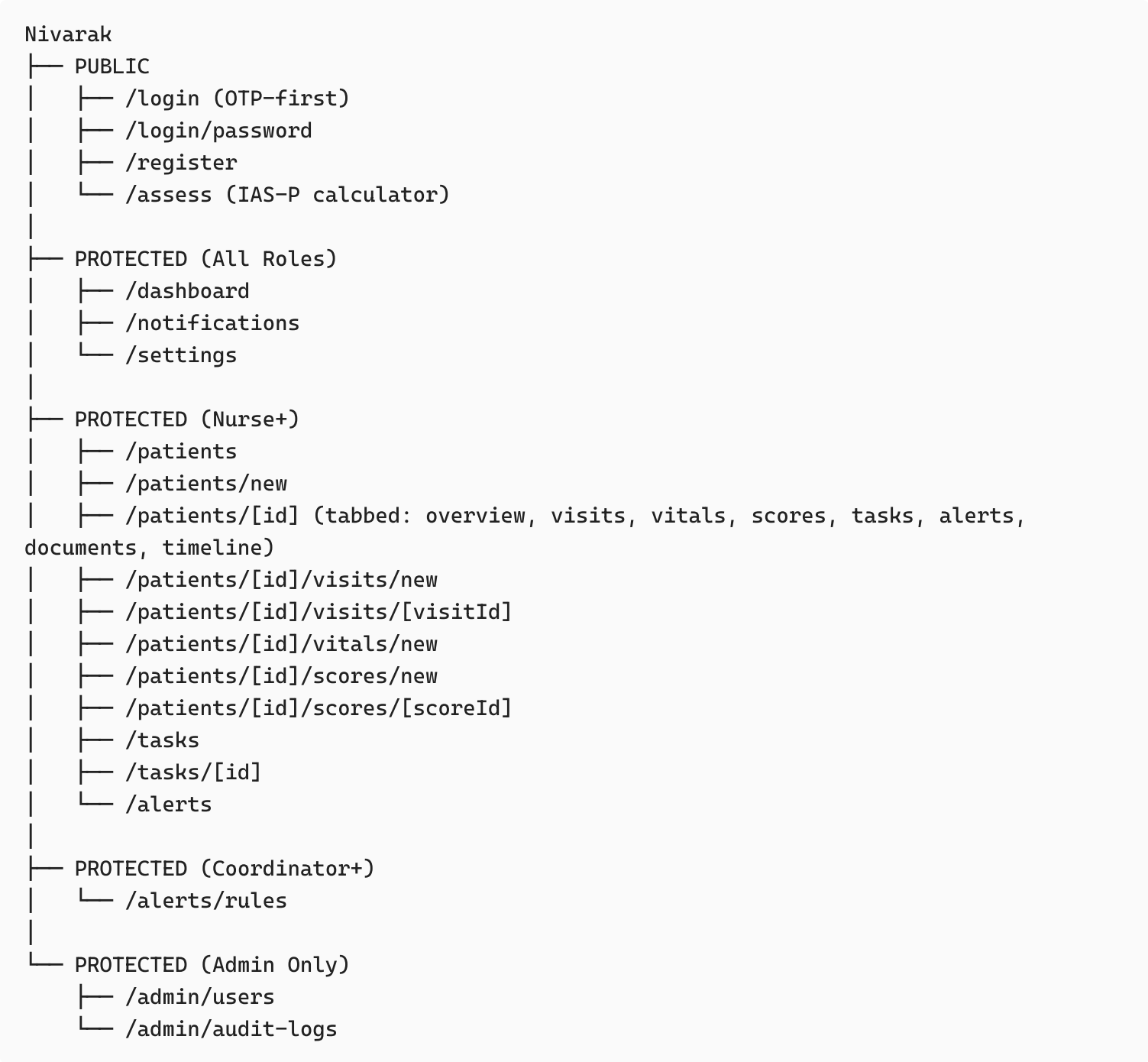
Band	Color	CSS Variable
Strong Independent (85-100%)	#22c55e (Green 500)	--risk-strong
Independent Vulnerable (70-84%)	#16a34a (Green 600)	--risk-independent
Supported Independence (55-69%)	#f59e0b (Amber 500)	--risk-supported
Limited Independence (40-54%)	#f97316 (Orange 500)	--risk-limited
High Dependence (0-39%)	#ef4444 (Red 500)	--risk-high

13.3 Design System Components (shadcn/ui)

Component	Variants	Usage
Button	default , destructive , outline , secondary , ghost , link + sizes: sm , default , lg , icon	All forms, actions, navigation
Input	default + error state	All text inputs
Textarea	default	Clinical notes, descriptions
Select	Single select + multi-select (via Command)	Filters, dropdowns
Checkbox	default	Red flags, bulk selection
RadioGroup	default	IAS-P assessment items (0/1/2 scale)
Switch	default	Settings toggles, theme
Dialog	default	Confirmations, create forms
Sheet	side: right	Create visit, record vital (slide-over panels)
Tabs	default	Patient detail tabs
Table	With DataTable wrapper	All list views
Card	default	KPIs, patient summary, vital readings
Badge	default , secondary , destructive , outline + custom risk/severity variants	Status labels throughout
Avatar	With fallback initials	User menu, patient header
Tooltip	default	Sidebar collapsed labels, info icons
Popover	default	Filter dropdowns
Calendar	With DatePicker	Date fields, date range
Accordion	default	FAQ sections, expandable details
Progress	default	Wizard progress bar, file upload
Skeleton	default	Loading states
ScrollArea	default	Sidebar, notification dropdown
DropdownMenu	default	User menu, row actions
Command	default	Global search, combobox
Separator	default	Visual dividers
Form	React Hook Form integration	All forms
Label	default	Form labels
Toast	Via Sonner	Success/error feedback

14. Deliverables Summary

14.1 Site Map

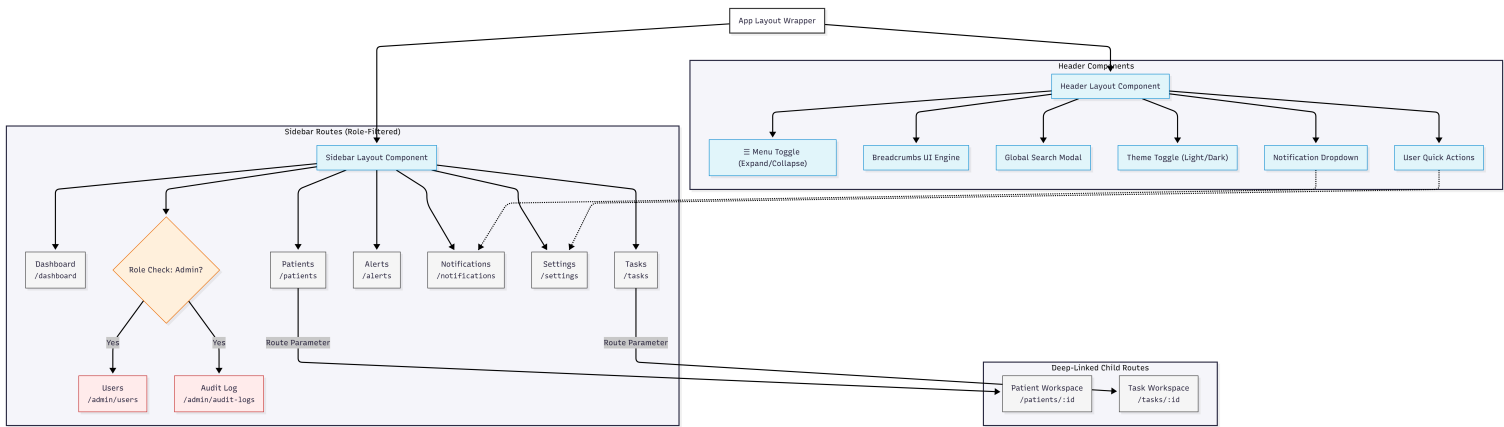


14.2 Route Map

Route	Page Component	Layout	Auth	Role Guard
/	Redirect	—	—	—
/login	LoginPage	AuthLayout	✗	—
/login/password	PasswordLoginPage	AuthLayout	✗	—
/register	RegisterPage	AuthLayout	✗	—
/assess	PublicAssessPage	PublicLayout	✗	—
/dashboard	DashboardPage	AppLayout	✓	Any
/patients	PatientListPage	AppLayout	✓	nurse+
/patients/new	CreatePatientPage	AppLayout	✓	nurse+
/patients/[id]	PatientDetailPage	AppLayout	✓	ABAC
/patients/[id]/visits/new	CreateVisitPage	AppLayout	✓	ABAC
/patients/[id]/visits/[visitId]	VisitDetailPage	AppLayout	✓	ABAC
/patients/[id]/vitals/new	RecordVitalPage	AppLayout	✓	ABAC
/patients/[id]/scores/new	AssessmentWizardPage	AppLayout	✓	ABAC
/patients/[id]/scores/[scoreId]	ScoreDetailPage	AppLayout	✓	ABAC

Route	Page Component	Layout	Auth	Role Guard
/tasks	TaskListPage	AppLayout	✓	nurse+
/tasks/[id]	TaskDetailPage	AppLayout	✓	nurse+
/alerts	AlertListPage	AppLayout	✓	nurse+
/alerts/rules	AlertRulesPage	AppLayout	✓	coordinator+
/notifications	NotificationsPage	AppLayout	✓	Any
/admin/users	AdminUsersPage	AppLayout	✓	admin
/admin/audit-logs	AuditLogsPage	AppLayout	✓	admin
/settings	SettingsPage	AppLayout	✓	Any

14.3 Navigation Flow Diagram



14.4 Architecture Summary

🔥 Important

Tech Stack: Next.js 14 (App Router) + shadcn/ui + Tailwind CSS + TanStack Query + Zustand + React Hook Form/Zod + Recharts + TanStack Table

Key Architecture Decisions:

- Route Groups:** Next.js App Router route groups (`auth`), (`public`), (`protected`) cleanly separate layout concerns
- Server State vs Client State:** TanStack Query owns all server data; Zustand only manages auth tokens, UI preferences, and notification badge count
- Feature-First Organization:** Components organized by feature domain (`features/patients/` , `features/scoring/`), not by component type
- Shared Component Library:** `components/ui/` (shadcn primitives) + `components/shared/` (domain components) + `components/charts/`
- API Service Layer:** Each backend module has a corresponding service file that encapsulates all HTTP calls
- Type Safety:** TypeScript types mirror the backend Zod schemas; Zod schemas are shared for form validation
- Role-Based UI:** Sidebar items, page access, and action buttons are filtered based on user role hierarchy (`admin` > `coordinator` > `doctor` > `nurse` > `caregiver` > `patient`)
- Polling for Real-Time:** Notifications and alerts use TanStack Query's `refetchInterval` for near-real-time updates (Phase 2: WebSockets)

Assumptions Made:

- A `GET /users` or `GET /auth/users` endpoint exists for admin user listing (not in HLD)
- A `GET /auth/me` endpoint exists for fetching current user profile (not explicitly in HLD, but middleware loads user context)

- 3. Patient edit uses PUT /patients/:id (exists in HLD)
- 4. The public IAS-P calculator and authenticated assessment wizard share the same form component
- 5. Document preview is basic (download link) — no inline PDF/image viewer in Phase 1

Component Count:

Category	Count
Pages	24
Feature Components	~85
Shared Components	~25
UI Primitives (shadcn)	~25
Chart Components	6
Custom Hooks	~17
API Service Files	12
Zustand Stores	3
Type Definition Files	12
Total Files (estimated)	~220